Subspaces

Subspace: a set $U \subseteq \mathbb{R}^n$ such that $\mathbf{0} \in U$, $\mathbf{u} + \mathbf{v} \in U$ for every $\mathbf{u}, \mathbf{v} \in U$, and $c\mathbf{u} \in U$ for every $c \in \mathbb{R}$ and $\mathbf{u} \in U$.

Closure under linear combinations: for a subspace U, if $\mathbf{u}_1, \dots, \mathbf{u}_k \in U$ and $c_1, \dots, c_k \in \mathbb{R}$, then $c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k \in U$.

Zero / trivial subspace: $\{\mathbf{0}\}$.

Proper subspace: a subspace U with $\{\mathbf{0}\} \subsetneq U \subsetneq \mathbb{R}^n$.

Null space and image space

Let A be an $m \times n$ real matrix, defining a map from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Null space: $\text{null}(A) = \{\mathbf{x} \in \mathbb{R}^n \,:\, A\mathbf{x} = \mathbf{0}\}$; input vectors sent to $\mathbf{0}$.

Image space: $\text{im}(A) = \{A\mathbf{x} \,:\, \mathbf{x} \in \mathbb{R}^n\}$; outputs $\mathbf{y} \in \mathbb{R}^m$ for which $A\mathbf{x} = \mathbf{y}$ has a solution.

Theorem: $\text{null}(A)$ is a subspace of $\mathbb{R}^n$; $\text{im}(A)$ is a subspace of $\mathbb{R}^m$.

Solution sets: $A\mathbf{x} = \mathbf{0}$ has solution set $\text{null}(A)$.

Spanning sets

Span: for $\mathbf{x}_1, \dots, \mathbf{x}_k \in \mathbb{R}^n$,

$$\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\} = \{t_1\mathbf{x}_1 + \cdots + t_k\mathbf{x}_k \,:\, t_1, \dots, t_k \in \mathbb{R}\}.$$

Spanning set: $\mathbf{x}_1, \dots, \mathbf{x}_k$ span V when $V = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$.

Theorem: $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is a subspace of $\mathbb{R}^n$ containing each $\mathbf{x}_i$. If a subspace $W \subseteq \mathbb{R}^n$ contains every $\mathbf{x}_i$, then $U \subseteq W$.

Columns span the image: for $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ and $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$,

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n, \qquad \text{im}(A) = \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}.$$

Basic solutions span the null space: the basic solutions of $A\mathbf{x} = \mathbf{0}$ span $\text{null}(A)$. With no free variables, $\text{null}(A) = \{\mathbf{0}\}$.